

COMPACTNESS AND INERTIA

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Let E be an elliptic curve over a local field K . The rational points of E with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and E has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 1.8 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 2.12 studies how the inertia group acts on the torsion of a curve with potential good reduction. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A finite extension K of $\mathbb{Q}_p$, with normalized valuation v , ring of integers $\mathcal{O}_K$, uniformizer π and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic p . (Silverman’s Chapter VII assumes that K and k are perfect, and here both are.)
- A fixed algebraic closure $\bar{K}$ of K .
- A bar denotes reduction modulo π .
- An elliptic curve E over K , given by a minimal Weierstrass equation $F(X, Y, Z) = 0$, with $F \in \mathcal{O}_K[X, Y, Z]$ homogeneous of degree 3. It embeds E in $\mathbb{P}^2$ as $V_+(F)$.
- The **primitive coordinates** $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every P has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The **reduction** $\bar{P} := [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve E_π , defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

How to use this. Allow about 25 minutes for each problem. The Magma block in Problem 2.12 runs in seconds in the free **online calculator**. Everything else is by hand.

1. FINITENESS BY COMPACTNESS

This section is [Sil09, Exercise 7.6]. The argument uses that k is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\bar{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman’s proof of it uses the finiteness of $E(K^{\text{unr}})/E_0(K^{\text{unr}})$, where K^{unr} is the maximal unramified extension of K , whose residue field is an algebraic closure of k [Sil09, proof of Theorem VII.7.1].

Definition 1.1 (The topology of K). We give K the topology defined by v : the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, K is a Hausdorff topological field.

Definition 1.2 (The subgroup $E_0(K)$). Let E_π^{ns} be E_π minus its singular point, if it has one. Then

$$E_0(K) := \{P \in E(K) : \bar{P} \in E_\pi^{\text{ns}}(k)\}.$$

Definition 1.3 (Regular functions). A function $\psi \in K(E)$ is **regular** at $Q \in E(\bar{K})$ if $\psi = g/h$ with $g, h \in K[X, Y, Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) := g(Q)/h(Q)$, an element of K if $Q \in E(K)$. The functions regular at Q form a discrete valuation ring with fraction field $K(E)$ [Poo06, Definition 1.8.3].

Proposition 1.4 ($E_0(K)$ is a subgroup). *The set $E_0(K)$ is a subgroup of $E(K)$ [Sil09, Proposition VII.2.1].*

Proposition 1.5 (Compactness). *The ring $\mathcal{O}_K$ is compact: it is complete, and for every n it is covered by the $(\#k)^n$ translates of $\pi^n \mathcal{O}_K$. Equivalently, K is locally compact [Poo17, §1.1.2].*

Proposition 1.6 (Morphisms $E \rightarrow E$ over K).

- (1) *The morphisms $\alpha: E \rightarrow E$ defined over K are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$ [Poo17, Proposition 3.6.5(b)].*
- (2) *If $Q \in E(\bar{K})$ and ψ_0, ψ_1, ψ_2 are regular at Q and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$ [Sil09, §I.3].*
- (3) *Replacing each ψ_i by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\bar{K})$ some ϕ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at Q and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 1.3 will do [Sil09, proof of Proposition II.2.1].*

Proposition 1.7 (Translations). *Let $P \in E(K)$. The translation $\tau_P: E \rightarrow E$, $Q \mapsto Q + P$, is an isomorphism defined over K , with inverse τ_{-P} [Sil09, Theorem III.3.6 and Example III.4.7].*

Problem 1.8 ($E(K)/E_0(K)$ is finite).

Let $q: K^3 - \{0\} \rightarrow \mathbb{P}^2(K)$ be the quotient map. Give $K^3 - \{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S := \{\mathbf{x} = (x_0, x_1, x_2) \in \mathcal{O}_K^3 : x_i \in \mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.

- (1) Show that S is compact and open in $K^3 - \{0\}$, and that $q(S) = \mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U) \cap S$ is open in S .^a
- (2) Show that $E(K) = V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.
- (3) Let $g, h \in K[X, Y, Z]$ be homogeneous of the same degree. Show that $W := \{Q \in \mathbb{P}^2(K) : h(Q) \neq 0\}$ is open, and that $W \rightarrow K$, $Q \mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 1.6 and 1.7 that every translation $\tau_P: E(K) \rightarrow E(K)$ is a homeomorphism.
- (4) Show that the reduction map $\mathbb{P}^2(K) \rightarrow \mathbb{P}^2(k)$, $P \mapsto \bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.
- (5) Prove that $E(K)/E_0(K)$ is finite.
- (6) Let K' be a field complete with respect to a discrete valuation whose residue field k' is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over K' ?

^aHint: $q^{-1}(U)$ is stable under multiplication by $K^\times$.

2. INERTIA ACTING ON TORSION

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geq 3$ added to its first two parts (item (5) explains why) and a Magma check. We use the notation of [Sil09, §VII.4].

Definition 2.1 (Reduction types). The curve E has **good reduction** if E_π is nonsingular, that is, if $v(\Delta) = 0$, and **bad reduction** otherwise. It has **potential good reduction** if it has good reduction over some finite extension of K .

Definition 2.2 (Inertia). Let $G_K := \text{Gal}(\overline{K}/K)$, $K^{\text{unr}} \subseteq \overline{K}$ the maximal unramified extension of K , and $I_v := \text{Gal}(\overline{K}/K^{\text{unr}})$ the **inertia group**. A set on which G_K acts is **unramified** if I_v acts trivially on it.

Definition 2.3 (Ramification of finite extensions). For an extension L of K inside $\overline{K}$, let $G_L := \text{Gal}(\overline{K}/L)$. If L/K is finite and Galois, the **inertia group** of L/K is the image of I_v in $\text{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap G_L)$. The extension L/K is **unramified** if its inertia group is trivial, and **tamely ramified** if its order is prime to p .

Definition 2.4 (Torsion fields and Tate modules). For $m \geq 1$ with $p \nmid m$, $K(E[m])$ is the field generated over K by the coordinates of the points of $E[m](\overline{K})$, so $G_{K(E[m])}$ is the kernel of the action of G_K on $E[m](\overline{K})$. For a prime $\ell \neq p$, the **Tate module** $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$ [Sil09, §III.7], and $\rho_\ell: G_K \rightarrow \text{Aut } T_\ell(E)$ is the action of G_K on it.

In the following results, $\ell \neq p$ is a prime and $m \geq 1$ is an integer with $p \nmid m$.

Proposition 2.5 (Minimal equations). *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal [Sil09, Remark VII.1.1]. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$ [Sil09, §III.1].*

Corollary 2.6 (Torsion). *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$ [Sil09, Corollary III.6.4(b)].*

Proposition 2.7 (The Tate module). *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$ [Sil09, Proposition III.7.1(a)]. Hence each projection $T_\ell(E) \rightarrow E[\ell^n](\overline{K})$ is a surjective homomorphism of G_K -modules, and its kernel is $\ell^n T_\ell(E)$, since both have index ℓ^{2n} .*

Proposition 2.8 (Good reduction). *If E has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified [Sil09, Proposition VII.4.1].*

Theorem 2.9 (Néron–Ogg–Shafarevich). *If $T_\ell(E)$ is unramified, then E has good reduction [Sil09, Theorem VII.7.1].*

Corollary 2.10 (Potential good reduction). *The curve E has potential good reduction if and only if $\rho_\ell(I_v)$ is finite [Sil09, Corollary VII.7.3].*

Proposition 2.11 (Inertia groups of finite extensions). *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension K' of K inside $\overline{K}$ whose inertia group $\text{Gal}(\overline{K}/K'K^{\text{unr}}) = I_v \cap G_{K'}$ is H [Sil09, proof of Corollary VII.7.3].*

Problem 2.12 (The inertia group of $K(E[m])/K$).

Suppose that E has potential good reduction, and let J be the subgroup of the elements of I_v that act trivially on $E[m](\overline{K})$ for every $m \geq 1$ with $p \nmid m$.

- (1) Let ℓ be a prime and $r \geq 1$ with $\ell^r \geq 3$, and let $g \in \text{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.^a
- (2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that J has finite index in I_v .
- (3) Let $m \geq 3$ with $p \nmid m$. Show that $J = I_v \cap G_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to I_v/J , independently of m .
- (4) Let $m \geq 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if E has good reduction. Which direction holds for every $m \geq 1$ with $p \nmid m$, and for every elliptic curve over K ?
- (5) Let $p \geq 3$ and $E: y^2 = x^3 - \pi^2 x$.
 - (a) Show that E has bad reduction, potential good reduction, and $K(E[2]) = K$.

- (b) Show that I_v/J has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as -1 .^b
- (c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geq 3$?
- (6) Suppose that $p \geq 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geq 1$ with $p \nmid m$.^c
- (7) Magma computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
    #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);           // 1
Inertia(-p^2, 0);         // 2
Inertia(0, p);            // 6
```

Check the first two values against your answers, and the third against (6).

^aHint: reduce to g of prime order q , write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not divisible by ℓ , and expand g^q .

^bHint: over $K(\sqrt{\pi})$, E is isomorphic to $y^2 = x^3 - x$.

^cHint: $\# \mathrm{GL}_2(\mathbb{F}_3) = 48$.

REFERENCES

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- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. [MR2514094](#) [↑](#)